



OGC

Open Standards Code Sprint

#Day 0 – Pre-event webinar

Joana Simoes

09/10/2025

What is OGC?



Our Vision

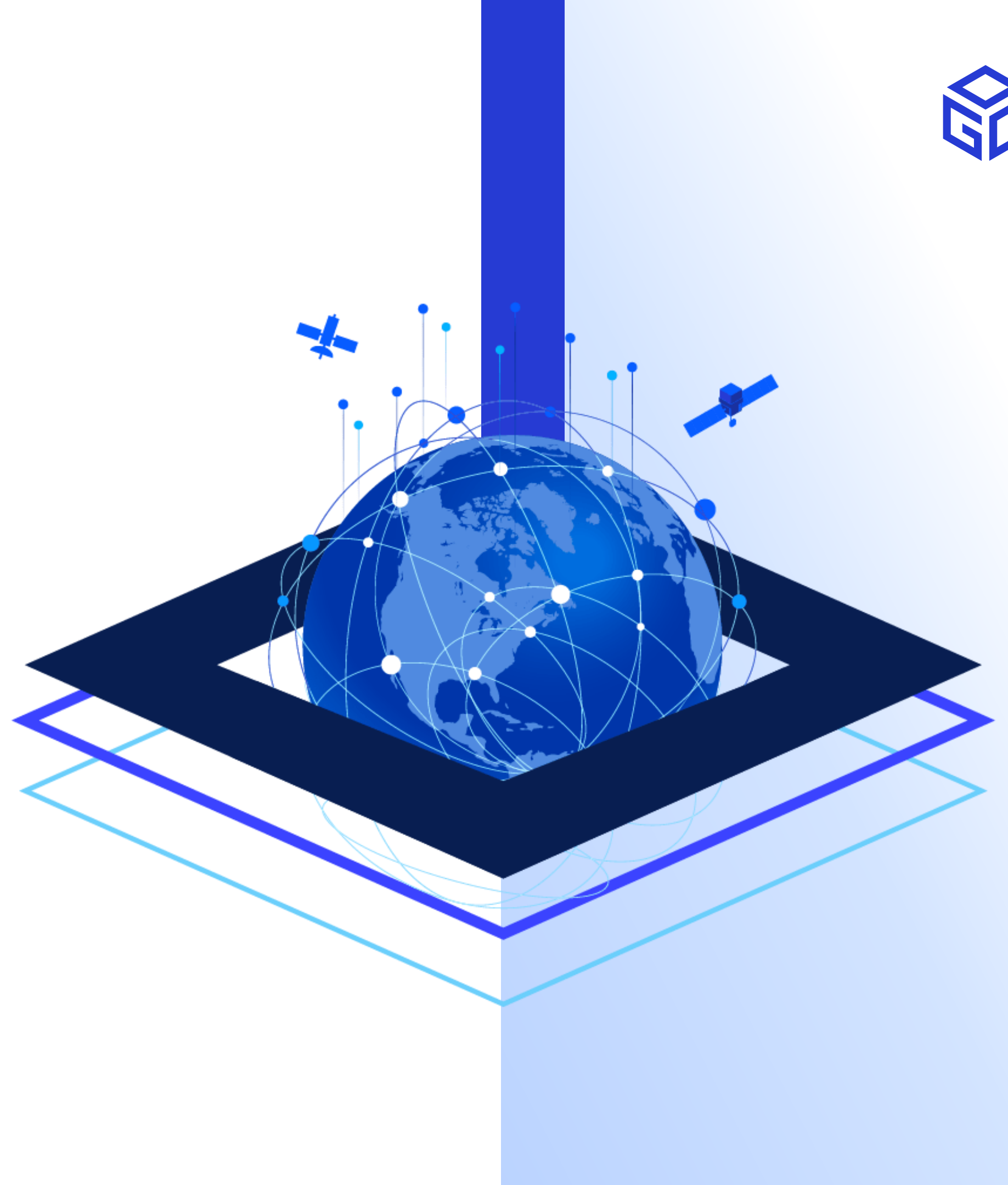
Building the future of location with community and technology for the good of society.

Our Mission

Make location Information Findable, Accessible, Interoperable and Reusable (FAIR)

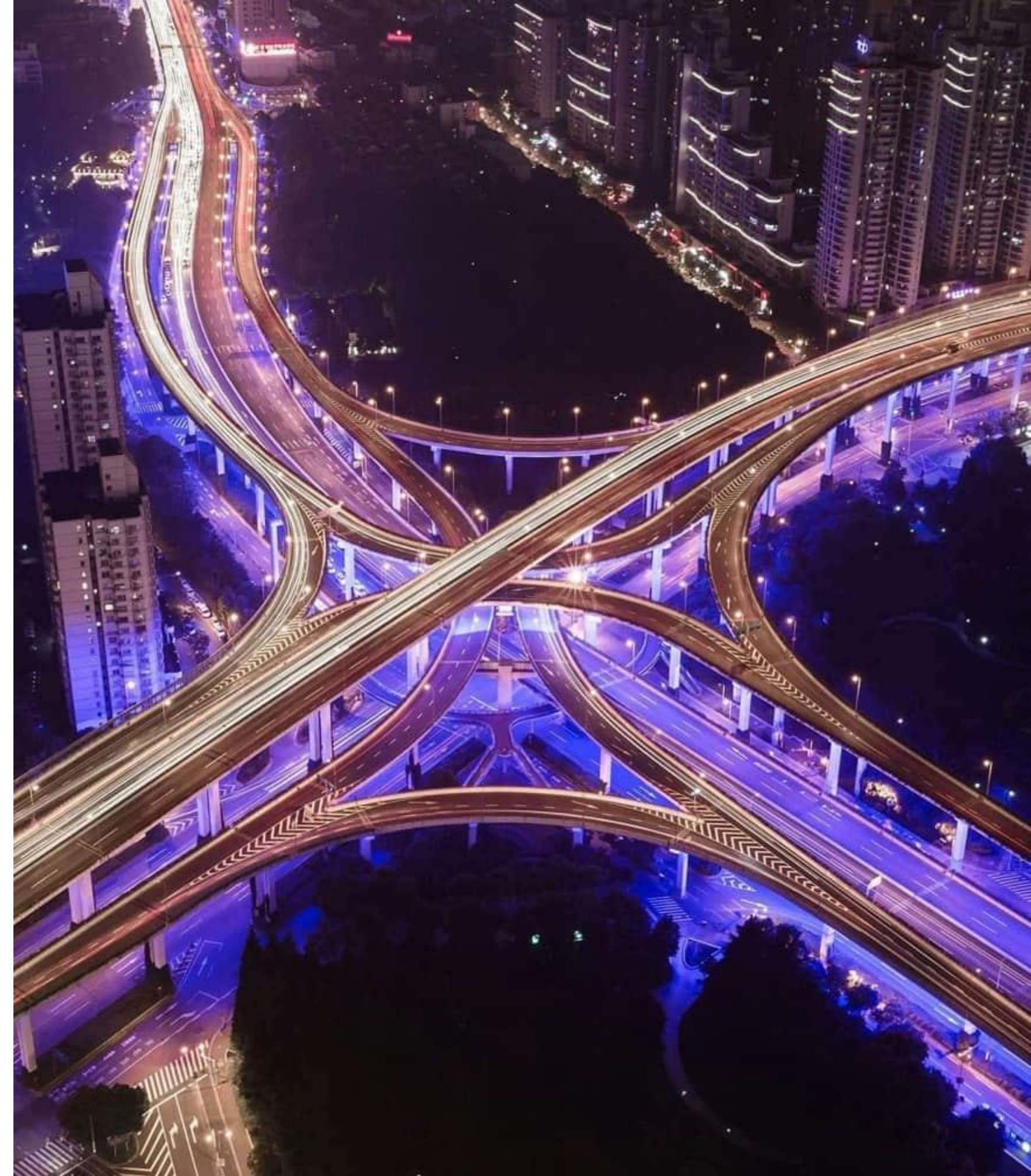
Our Approach

A proven collaborative and agile process combining consensus-based standards, innovation project and partnership building.

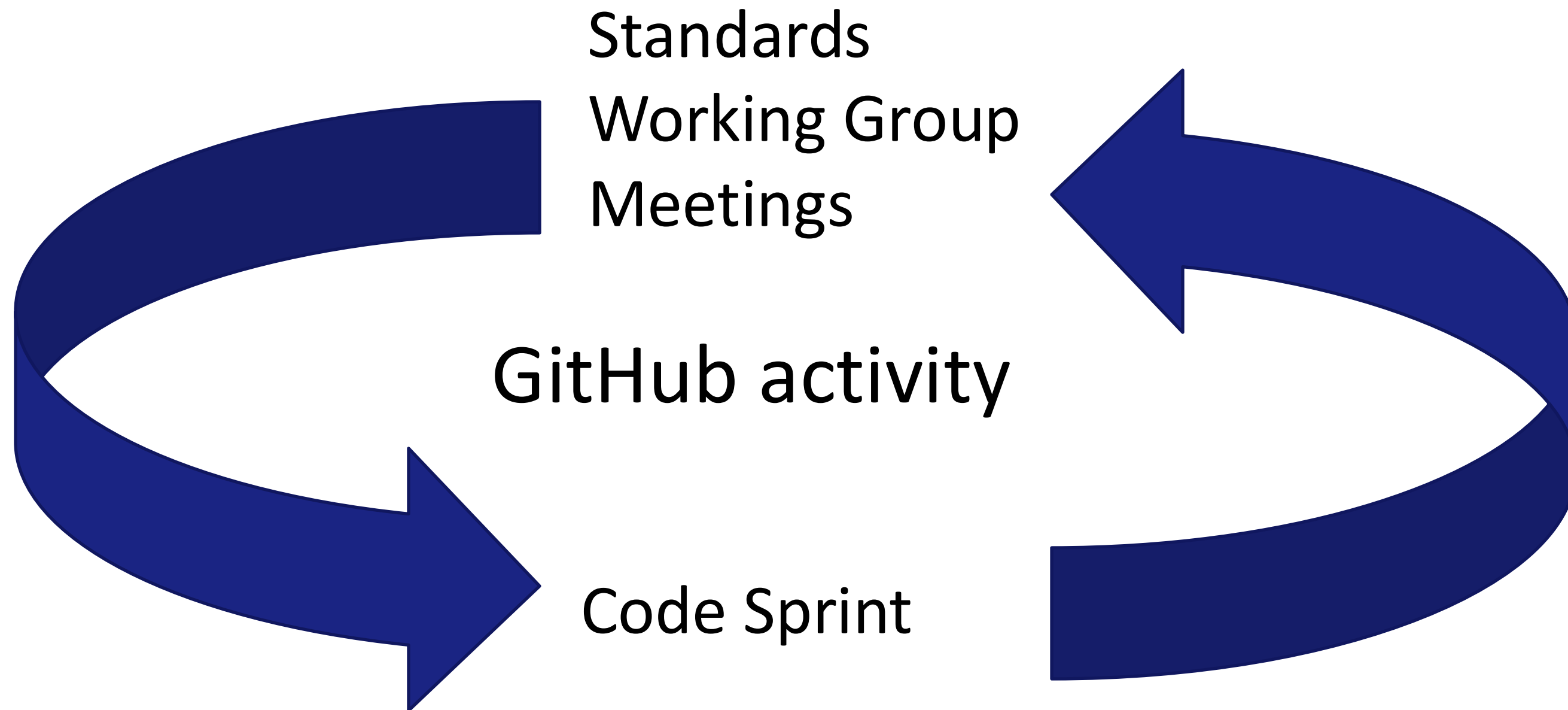


Some advantages of adopting Standards

- Server side: enable a wide range of clients to consume services (e.g.: no need to create custom clients).
- Client side: being able to consume services from a wide range of servers (e.g.: add support to more sources with minimal coding).
- More data access, less coding.



OGC Code Sprints



- Support the development of the standards.
- Support the development of implementations of the standards.
- Create awareness about the standards.

Thank You to Our Hosts!



Gemeente
Rotterdam

Vonk Innovation Center

The image shows a web browser window with two tabs. The active tab is titled "OGC Code Sprint Nodes" and displays the URL `developer.ogc.org/sprints/nodes/`. The page content is a map of Rotterdam, with a blue pin marking the location of the "VONK Innovation Center". A tooltip above the pin shows the name and a link to "More info".

Overlaid on the right side of the browser window is a raw JSON file viewer. The URL bar shows `raw.githubusercontent.com/ogcincubator/nodes/refs/heads/main/src/poi.json`. The JSON content is as follows:

```
215     "hasPayload": {
216       "usesSchema": [
217         {
218           "href":
219             "https://raw.githubusercontent.com/ogcincubator/nodes/refs/heads/main/src/poi.json",
220           "rel": "describedby"
221         }
222       ],
223       "onlineResources": [
224         {
225           "description": "Website about organisation",
226           "linkage": "https://www.vonkrotterdam.nl/"
227         }
228       ],
229       "locationInfo": [
230         {
231           "address": {
232             "country": "Netherlands",
233             "deliveryPoint": "Het Timmerhuis, Halvemaanpassage
234             90, 3011 DL Rotterdam, Nederland"
235           },
236           "phone": {
237             "number": "NA"
238           }
239         }
240       ]
241     }
```

Logistics Info (in-person)



- Timmerhuis - Halvemaanpassage 1 or Halvemaanpassage 90 or Rodezand 14, Rotterdam
- Auditorium, VONK and Kapsalon rooms

What happens during the code sprints?



- Discussions



- Independent Work



- Mentor Stream

Tutorials – Onboarding Day

Mentor Stream Onboarding



@Mentor Room: 10:10 - 10:30 UTC+2



Open to All OGC standards

Standards Working Groups that have registered for this code sprint:

Joana Simoes (OGC), discord: @doublebyte.

Organising Committee

Gobe Hobona (OGC), Scott Simmons (OGC), and Linda van den Brink (Geonovum).

Register

Registration is free for everyone. You may register [here](#). At the time of the event, only registered participants will have access to the relevant Discord channels.

Which OGC standards working groups are going to participate?

- [OGC API - Processes](#)
- [OGC SensorThings API](#)
- [OGC API - Connected Systems](#)
- [OGC API - Joins](#)
- [OGC API - Records](#)
- [OGC Pub/Sub](#)
- [OGC API - Coverages](#)
- [OGC API - DGGS](#)
- [Cartographic Symbolology](#)
- [JSON-FG](#)

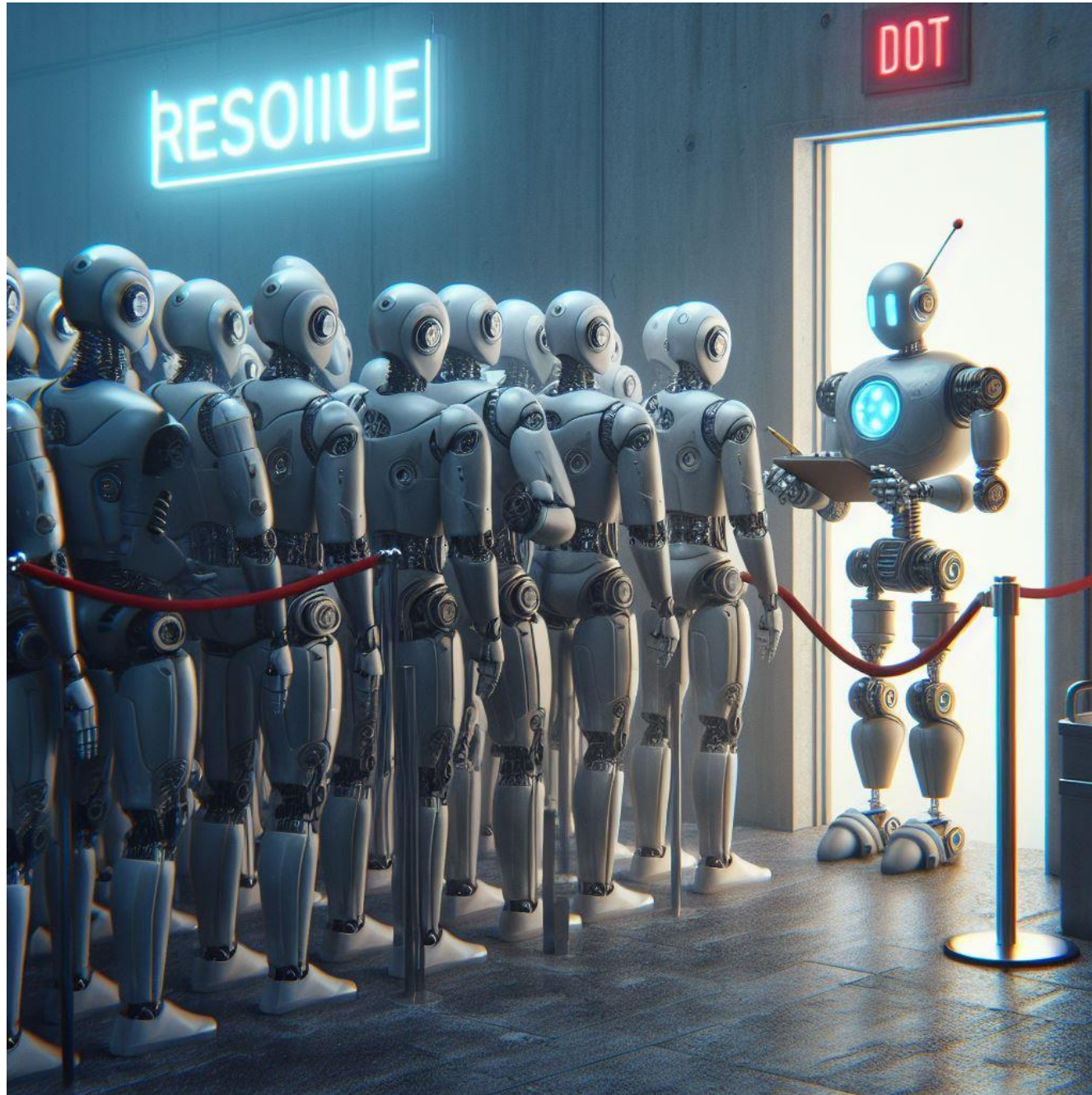
Special Tracks

Validators

In previous codesprints a linter was developed for JSON-FG, and expanded to support various OGC API Standards: <https://github.com/Geonovum-labs/ogc-checker>

Add Standards Working Group here

Special Tracks



- Validators



- DCAT and OGC API - Records



Overviews and Sprint Goals





OGC API - Processes, Records and JSON-FG





OGC API – Connected Systems



Sprint Goals

- **Confirm that the OAS definitions work as intended and can be used to autogenerate useful code without issues**
- **Confirm that a generic API – Features client (such as QGIS) can connect to a CSAPI server as expected without issues**
- ➔ • **Arrive at three or more stable, independent open-source implementations of CSAPI**

- Additional Goals to Refine:
 - Conformance Test development for API – Connected Systems Parts 1 and 2 (and encodings?)
 - Prototyping and testing OGC API – Connected Systems with OGC Pub/Sub using Web Video Map Tracks (WebVMT) as a use case

OS4CSAPI Initiative

POC, Description, & Links:

- POC – Sam Bolling, sbolling@riversideresearch.org
- A pilot effort to identify, engage, and assist open-source projects of interest with implementation of the OGC CSAPI v1.0 standard

Work Plan:

- **Draft actionable goals per project**
- **Draft actionable work items per project**
- **Prioritize Projects, Goals, and Work Items**
- **Assign Resources to Tasks**
- Execute Plan
- Report Results
- Discuss Future Work and Way Forward

Accomplishments & Status:

- Identified and engaged seven open-source repositories of interest
- GitHub Project Board created, updated, organized, and issues linked
- Camp-To-Camp - Fork, Implementation Plan, and Issues created
- 52 North – issues created and labeled for the sprint
- OWS Lib – issues created for the sprint

Challenges & Risks:

- **Developer Resources available**
- Alignment of Developer Resources to Tasks
- Time and Opportunity to Work on Tasks

Repositories of Interest

Project	Description	Open	Client/Server	Language	Response?	Engaged?	CSAPI P1	CSAPI P2
Open Sensor Hub	Java-based middleware for building better Sensor Webs and the Internet of Things	Yes	Server	Java	No	Yes		
52 North CSAPI PyGeoAPI	Proof of Concept of OGC Connected Systems API based on pygeoapi	Yes	Server	Python	Yes	Yes		
Botts Innovation CSAPI	Connected Systems API for Java	Yes	Server	Java	No	Yes		
GeoServer	GeoServer is an open-source software server written in Java that allows users to share and edit geospatial data. Designed for interoperability, it publishes data from any major spatial data source using open standards.	Yes	Server	Java		No		
HAKUNAPI	OGC API Features Java server	Yes	Server	Java		Yes	Planned	Not Planned
MERLIN (Server)	MERLIN is a reference implementation of standards of interest from the National MASINT Standards Baseline	No	Server	Java	Yes	Yes	Planned	Planned
QGIS (Plugin)	QGIS is a free, open source, cross platform (lin/win/mac) geographical information system (GIS)	Yes	Client	C++, Python	Yes	Yes		
OWSLib	OWSLib is a Python package for client programming with Open Geospatial Consortium (OGC) web service (hence OWS) interface standards, and their related content models.	Yes	Client	Python	Yes	Yes		
Camp to Camp	A TypeScript library for interacting with geospatial services	Yes	Client	TypeScript	Yes	Yes		
MERLIN (Client)	MERLIN is a reference implementation of standards of interest from the National MASINT Standards Baseline	No	Client	TypeScript	Yes	Yes	Planned	Planned

OS4CSAPI GitHub Project Board



@OS4CSAPI

Short description

Open Source for CSAPI (OS4CSAPI) is a project that coordinates the open source adoption of the OGC API – Connected Systems v1 standard across multiple GitHub projects. Sprint goals include validating OpenAPI definitions, testing with OGC API - Features clients, and achieving stable implementations.

README

OS4CSAPI Initiative

Welcome to the coordination space for the **Open Source for Connected Systems API (OS4CSAPI)** pilot project. This initiative connects open source geospatial projects with the [OGC API – Connected Systems v1](#) standard to accelerate adoption and implementation.

Project Goals

- ✓ Validate OpenAPI definitions for CSAPI Parts 1 & 2
- ✓ Ensure client tools (e.g., QGIS) can connect to CSAPI servers
- ✓ Achieve 3+ stable, independent open-source implementations
- 🟡 Begin conformance test development
- 🟡 Prototype CSAPI with [OGC Pub/Sub](#)

Backlog Table · @OS4CSAPI

github.com

Sam-Bolling / Projects / @OS4CSAPI

@OS4CSAPI

Kanban Board Backlog Table + New view

Filter by keyword or by field

Repository	Title	Assignees	Status	Labels	Epic
1 geopython/OWSLib	CSAPI tests failing #1013		Backlog	OGC API	Stable Implementations
2 geopython/OWSLib	Support for OGC API - Connected Systems #1011		Backlog		Stable Implementations
3 qgis/QGIS	Support for OGC API - Connected Systems #63108		Backlog	Feature Request Server	Stable Implementations
4 52North/connected-systems	[BugFix] Small Clean ups to get local dev version #9		Backlog		Stable Implementations
5 52North/connected-systems	Update pygeoapi dependency #11		Backlog	enhancement good first issue	Stable Implementations
6 52North/connected-systems	Identify work items for OGC Code Sprint 26 #8	SpeckiJ	Backlog	enhancement	Stable Implementations
7 52North/connected-systems	Document Relation to pygeoapi #10	SpeckiJ	Backlog	documentation	Stable Implementations
8 nlsfi/hakunapi	Support for OGC API - Connected Systems #153		Backlog		Stable Implementations
9 opensensorhub/osh-connector	Identify work items for OGC Code Sprint 26 #313		Backlog		Stable Implementations
10 Botts-Innovative-Research/ConnectedSystemsAPI-Java	Identify work items for OGC Code Sprint 26 #1		Backlog		Stable Implementations
11 camptocamp/ogc-client	Support for OGC API - Connected Systems #118		Backlog		Stable Implementations
12 Sam-Bolling/ogc-client	[Work Item] Scaffold CSAPI module structure #1		Backlog	architecture setup	Stable Implementations
13 Sam-Bolling/ogc-client	[Work Item] Implement get_systems() method #2		Backlog	api-methods feature	Stable Implementations
14 Sam-Bolling/ogc-client	[Work Item] Implement get_observations() method #3		Backlog	api-methods feature	Stable Implementations
15 Sam-Bolling/ogc-client	[Work Item] Implement get_system_history() method #4		Backlog	api-methods feature	Stable Implementations
16 Sam-Bolling/ogc-client	[Work Item] Implement get_commands() method #5		Backlog	api-methods feature	Stable Implementations
17 Sam-Bolling/ogc-client	[Work Item] Add capability detection for CSAPI #6		Backlog	architecture feature	Stable Implementations
18 Sam-Bolling/ogc-client	[Work Item] Integrate CSAPI into main client #7		Backlog	architecture integration	Stable Implementations

Linked Projects

We're collaborating with the following open source projects:

- <https://github.com/opensensorhub/osh-connector>
- [52North/connected-systems-pygeoapi](https://github.com/52North/connected-systems-pygeoapi)
- [geopython/OWSLib](https://github.com/geopython/OWSLib)
- [camptocamp/ogc-client](https://github.com/camptocamp/ogc-client)
- [qgis/QGIS](https://github.com/qgis/QGIS)
- [nlsfi/hakunapi](https://github.com/nlsfi/hakunapi)
- <https://github.com/Botts-Innovative-Research/ConnectedSystemsAPI-Java>

17 Sprint Planning

Check out our [GitHub Project Board](#) for current sprint goals, tasks, and progress.

How to Contribute

- Join the conversation in linked issues
- Suggest new projects or use cases
- Help test, prototype, or document CSAPI integrations

Contact

For questions, feedback, or collaboration ideas related to OS4CSAPI:

- Comment on any of the linked GitHub issues in participating repositories
- Reach out via GitHub (mention me in an issue or discussion)
- Follow updates and progress on the OS4CSAPI Project Board

OS4CSAPI Initiative – 52 North CSAPI PyGeoAPI

POC, Description, & Links:

- POCs - [Organizational] Simon Jirka (jirka@52North.org), [Technical] Jan Speckamp (j.speckamp@52North.org), [Technical] Christian Autermann (c.autermann@52North.org)
- 52°North is developing a Server Implementation of the OGC CSAPI Standard in python. Core feature is the seamless integration with the pygeoapi Project, allowing for co-hosting CSAPI with other OGC API Family standards. Currently the project is in alpha-phase, the first production-ready release is planned for Q2 2026.
- <https://github.com/52North/connected-systems-pygeoapi>
- Sprint Goal – contribute to the addition of an OGC CSAPI v1.0 implementation to this project

Accomplishments & Status:

- Project Engaged and Responded
- <https://github.com/52North/connected-systems-pygeoapi/issues/8>
- Project Maintainer provided metadata and additional support
- Project Maintainer drafted initial, high-level goals and work plan

Work Plan:

- Identify which Sprint Idea is worked on during the sprint
- Develop+Publish (Long-Term) Roadmap
- Add non-technical documentation
- Showcase how to use the standard to store data from client-view
- Sprint Ideas:
 - Test Specification Conformance
 - Develop suite of integration-data for Part 1 and 2
 - Structure test data in a way to be reusable by official ETS
 - Implement Integration with Generic Features Clients (QGIS + OWSLib)
 - Currently not working out-of-the-box
 - Document Best-Practice for Backend Developers
 - Demonstrator for integration with pygeoapi
 - "pygeoapi" is part of the Repo name, but no documentation on the actual integration
 - Setup Demo highlighting cohosting of CSAPI data with other OGC API Family Specifications
 - Prototype CSAPI Part 3: Pub/Sub
 - Create base for easily testing out ideas during specification development
 - Learning from Part 1 and 2

Challenges & Risks:

- Support from 52°North limited by restricted resources

OS4CSAPI Initiative – Camp to Camp

POC, Description, & Links:

- POC - **TBD**
- The camp-to-camp OGC-Client is a TypeScript library for interacting with geospatial services. It implements several OGC standards and will help you interact with them in a user-friendly and consistent way.
- [camptocamp/ogc-client: A TypeScript library for interacting with geospatial services](#)
- Sprint Goal – contribute to the addition of an OGC CSAPI v1.0 implementation to this project

Accomplishments & Status:

- Project Engaged and Responded
- Project forked, implementation plan created, work items created
 - Forked Project with plan and work items:
 - [Sam-Bolling/ogc-client: Fork of camptocamp/ogc-client to implement OGC API Connected Systems support \(CSAPI\), including modular client, tests, and documentation.](#)
- Upstream Repository issue:
 - [Support for OGC API - Connected Systems · Issue #118 · camptocamp/ogc-client](#)

Work Plan:

- **Draft actionable goals**
- **Draft actionable work items**
- **Prioritize Goals and Work Items**
- **Assign Resources to Tasks**
- Execute Plan
- Report Results
- Discuss Future Work and Way Forward

Challenges & Risks:

- **Developer Resources available**



OS4CSAPI Initiative – OWSLib

POC, Description, & Links:

- POC - **TBD**
- OWSLib is a Python package for client programming with Open Geospatial Consortium (OGC) web service (hence OWS) interface standards, and their related content models.
- <https://github.com/geopython/OWSLib>
- Sprint Goals:
 - Assess any changes between the draft and final specification, and update OWSLib accordingly
 - Fix issues causing CSAPI unit tests to fail

Work Plan:

- **Draft actionable goals**
- **Draft actionable work items**
- **Prioritize Goals and Work Items**
- **Assign Resources to Tasks**
- Execute Plan
- Report Results
- Discuss Future Work and Way Forward

Accomplishments & Status:

- Project Engaged and Responded
- [Support for OGC API - Connected Systems · Issue #1011 · geopython/OWSLib](#)
- [CSAPI tests failing · Issue #1013 · geopython/OWSLib](#)
- [OGC Connected Systems API by ChainReaction31 · Pull Request #928 · geopython/OWSLib](#)

Challenges & Risks:

- **Developer Resources available**

geographika 19 hours ago

Contributor

@Sam-Bolling - OWSLib implemented support for the draft specs of "OGC API - Connected Systems - Part 1: Feature Resources" and "OGC API - Connected Systems - Part 2: Dynamic Data" in 2024. See:

- <https://owslib.readthedocs.io/en/latest/features.html#ogc-api-support>
- [Code example](#)
- Implementation provided in [OGC Connected Systems API #928](#) by **@ChainReaction31**

It would be good to assess any changes between the draft and final specification, and update OWSLib accordingly.

tomkralidis opened 6 hours ago

As per [#1012 \(comment\)](#), looks like the server in the unit tests is offline.

@ChainReaction31 can you help here (given [#928](#))? Thanks (cc **@Sam-Bolling**).

OS4CSAPI Initiative – QGIS

POC, Description, & Links:

- POC - **TBD**
- QGIS is a free, open source, cross platform (lin/win/mac) geographical information system (GIS)
- [qgis/QGIS: QGIS is a free, open source, cross platform \(lin/win/mac\) geographical information system \(GIS\)](#)
- Sprint Goal – contribute to the addition of an OGC CSAPI v1.0 implementation to this project

Work Plan:

- **Draft actionable goals**
- **Draft actionable work items**
- **Prioritize Goals and Work Items**
- **Assign Resources to Tasks**
- Execute Plan
- Report Results
- Discuss Future Work and Way Forward

Accomplishments & Status:

- Project Engaged and Responded
- [Support for OGC API - Connected Systems · Issue #63108 · qgis/QGIS](#)

Challenges & Risks:

- **Developer Resources available**



OS4CSAPI Initiative – HAKUNAPI

POC, Description, & Links:

- POC - **TBD**
- HAKUNAPI is an OGC API Features Java server
- [nlsfi/hakunapi: OGC API Features Java server](#)
- Sprint Goal – contribute to the addition of an OGC CSAPI v1.0 implementation to this project

Work Plan:

- **Draft actionable goals**
- **Draft actionable work items**
- **Prioritize Goals and Work Items**
- **Assign Resources to Tasks**
- Execute Plan
- Report Results
- Discuss Future Work and Way Forward

Accomplishments & Status:

- Project Engaged and Responded
- [Support for OGC API - Connected Systems · Issue #153 · nlsfi/hakunapi](#)

Challenges & Risks:

- **Developer Resources available**



Attendance Planning and Useful Links

SWG Member	26 Code Sprint
Mr. Jim Antonisse	
Mr. Dave Archer	Virtual
Mr. Sam Bolling	Virtual
Dr. Mike Botts	Tentative In-Person / Virtual
Mr. Drew Botts	Tentative In-Person / Virtual
Mr. Glenn Laughlin	Virtual
Mr. Eric LoMonaco	Virtual
Mr. Tom Kralidis	Tentative In-Person
Mr. Alex Robin	Tentative In-Person / Virtual
Mr. Rob Smith	
Mr. Jan Speckamp	In-Person
Dr. Chris Tucker	Tentative In-Person / Virtual

Links:

- [Open Standards Code Sprint](#)
- [October 2025 Open Standards Code Sprint · opengeospatial/developer-events Wiki](#)
- [What is everybody going to be working on, at the October 2025 Open Standards Sprint? · Issue #173 · opengeospatial/developer-events](#)
- [OGC Events Discord server](#)
- [Backlog Table · @OS4CSAPI](#)



OGC API – Joins





OGC API – Coverages, DGGS and Cartographic Symbolology





DCAT and OGC API - Records





Linters and Validators



More Information

GitHub repo:

<https://github.com/opengeospatial/developer-events/>

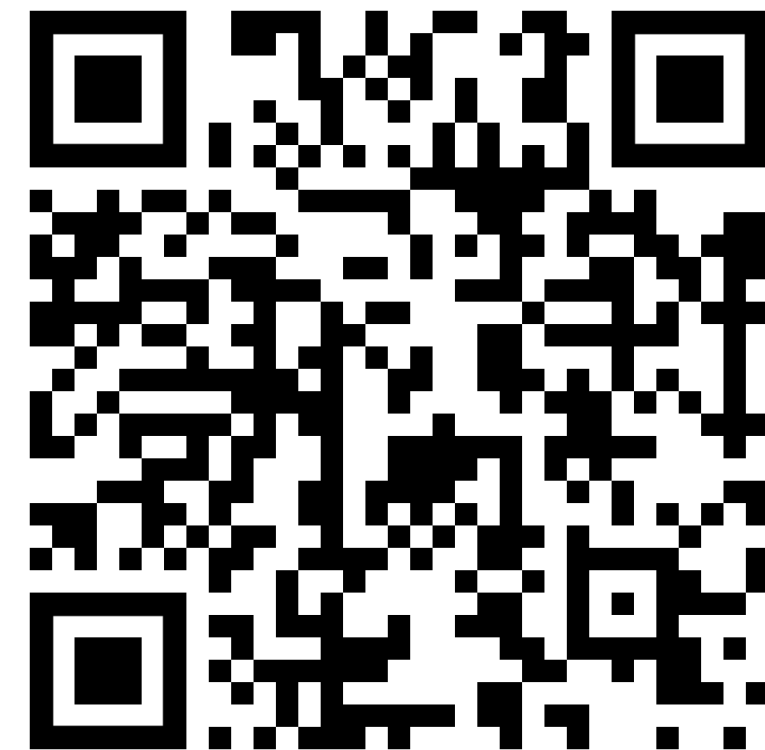
Folder 📁 2025/October-Code-Sprint

Wiki page:

<https://github.com/opengeospatial/developer-events/wiki/October-2025-Open-Standards-Code-Sprint>

Developer website:

<https://developer.ogc.org>



Hashtags: #CODESPRINT26 #OGCAPI



#173

Share what you will be working during the sprint

The screenshot shows a web browser window displaying a GitHub issue page. The browser's address bar shows the URL `github.com/opengeospatial/developer-events/issues/173`. The issue title is "What is everybody going to be working on, at the October 2025 Open Standards Sprint? #173". Below the title, there is a green "Open" button. The issue was opened by user "doublebyte1" on August 6. The main content of the issue is a text box with the placeholder "Post a comment letting us know what you are going to be working on." and a "Create sub-issue" button. Below this, there is a comment from user "securedimensions" dated August 7, which describes the "STA-WebSub" extension to the SensorThings API v1.1. The right sidebar of the issue page shows various metadata fields: Assignees (No one - Assign yourself), Labels (No labels), Type (No type), Projects (No projects), Milestone (No milestone), and Relationships (None yet).

What is everybody going to be working on, at the October 2025 Open Standards Sprint? #173

Open

doublebyte1 opened on Aug 6

Post a comment letting us know what you are going to be working on.

Create sub-issue

securedimensions on Aug 7 · edited by securedimensions

STA-WebSub

STA-WebSub defines an extension to SensorThings API v1.1 that allows clients to establish asynchronous messaging, over HTTP, based on self defined and discovered topics. The STA-WebSub extension is currently an [OGC draft Standard](#).

Related SWGs

The SensorThings API is the SWG that is currently working on the STA-WebSub extension.

Assignees: No one - [Assign yourself](#)

Labels: No labels

Type: No type

Projects: No projects

Milestone: No milestone

Relationships: None yet

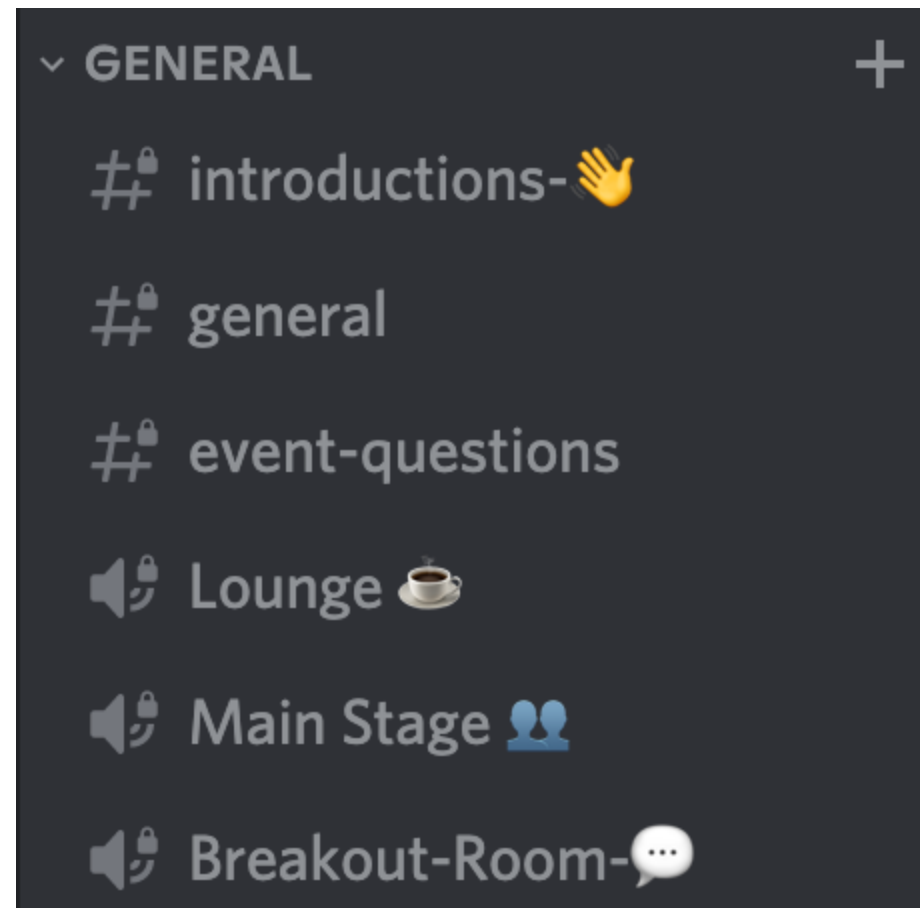
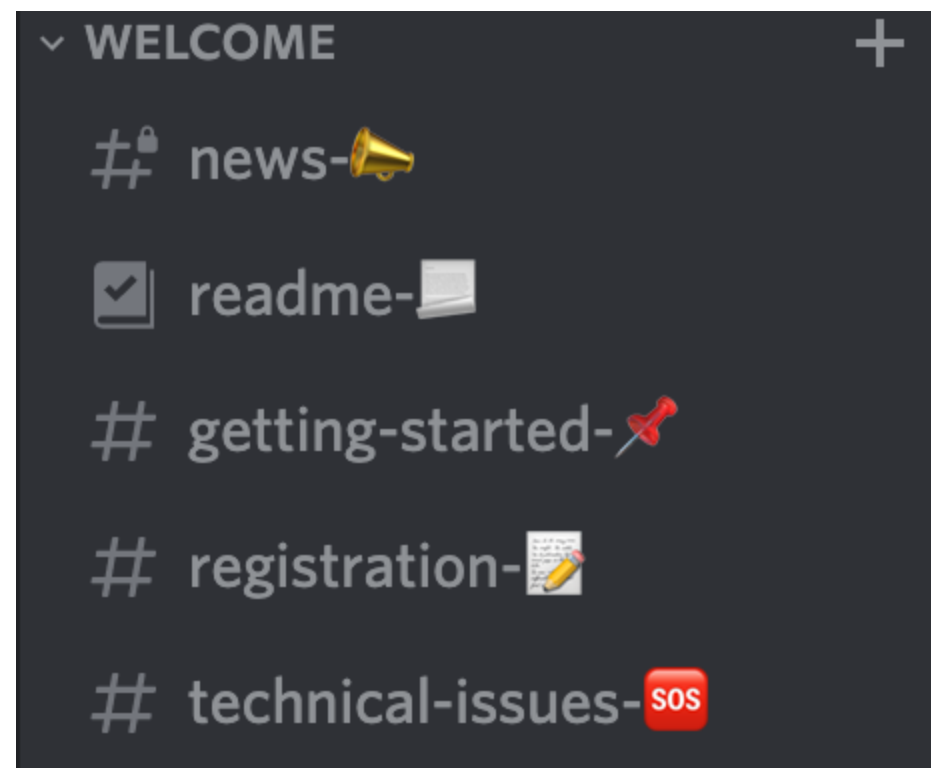
<https://github.com/opengeospatial/developer-events/issues/173>



OGC Events Discord Server



Meta channels:
check-in and
general info



Main track: Kick-off, standups and demos

Mentor stream:
tutorials, 1:1
mentoring



Matrix bridge: <https://matrix.to/#/%23ogc:matrix.org>





THANK YOU

**See you, virtually or in-person, on
Monday, October 20, at 9am UTC+2!**

www.ogc.org

